“ EDGE EYE “

AN AI BASED BLIND SPOT HUMAN DETECTION WITH AUTO BRAKING SYSTEM

Name of Project:
“EDGE EYE”

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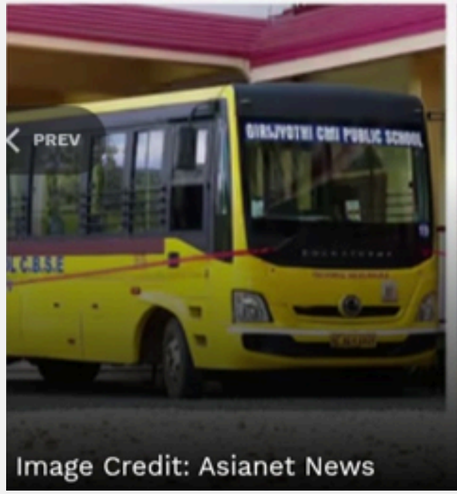

Image Credit: Asianet News

78-yr-old woman hit by school bus, dies

Dwaipayan Ghosh / Feb 24, 2025, 23:11 IST

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Kolkata: An avid morning walker (78), lost her life after a school bus hit her barely a few metres from the Wellington crossing around 7.10 am on Monday. The police said the victim, identified as Shanti Devi Singh, a resident of Rani Rashmoni Road in the Janbazar area, might have fallen in the blind spot of the driver. However, cops are still not clear about the exact circumstances.

The school bus, carrying over 20 school children studying at two reputed missionary schools on AJC Bose Road, sped away immediately, said Sealdah guard officials. The driver stopped the bus near Entally Market, close to the schools, and asked the attendant to take care of the children. He then fled. The police traced the bus driver and arrested him. The local police station claimed the bus had dropped the children at a missionary girls' school located nearby and was taking a left turn when the accident took place.

According to Rajkishore Singh, an executive with a private firm and the victim's son, he and his mother had the regular habit of going for morning walks. "On Monday, we went in our separate directions, my mother choosing to walk towards Wellington Square," said Singh.

Cops said the school bus was waiting to pick up a school student when Shanti might have decided to cross over. "What she might not have realised was that she was too close to the bus. When the school bus began to move, the driver did not spot the woman and the accident took place," said an officer.

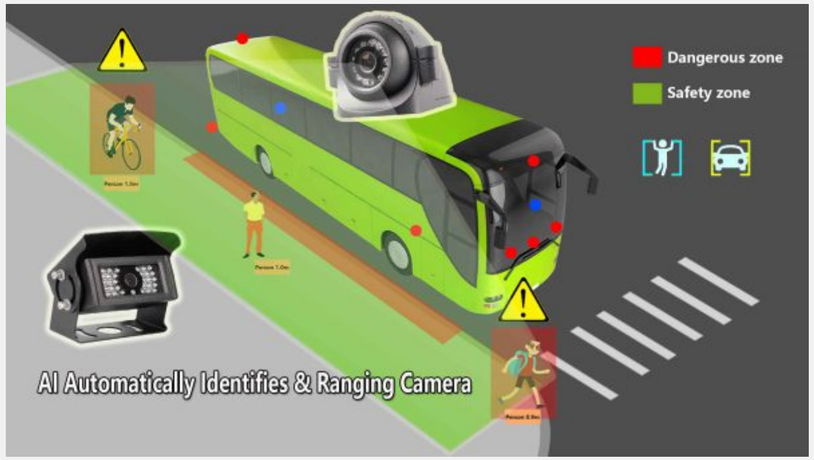
Singh told TOI that another woman who was at the spot was the first to return to Janbazar and inform of the accident. "By the time I reached the accident spot, my mother was shifted to NRS Hospital.

INTRODUCTION

- Child safety around school buses is a serious concern due to large blind spots.
- Drivers may not always see children standing or moving near the bus.
- This can lead to dangerous accidents .
- The project uses Artificial Intelligence to detect children in blind spot areas.
- When a child is detected, an LED indicates braking action instead of applying real brakes.
- The system is designed as a safe and low-cost prototype to demonstrate intelligent vehicle safety.

PROBLEM STATEMENT

School buses have large blind spots that can cause accidents involving children near the vehicle. Relying only on the driver’s attention is not always safe, and implementing real automatic braking in a student project is risky and complex. Therefore, the challenge is to design a safe, low-cost AI system that detects children in blind spots and demonstrates braking decisions without physically controlling the brakes.



OBJECTIVES

- To developavision-based system fordetecting children in vehicle blind spots.
- To use a machine learning object detection model for real-time human detection.
- To define a danger zone where detection triggers safety action.
- To simulate automatic braking commands using an LED indicator.
- To provide visual and audio alerts to assist the driver.
- To create a safe prototype suitable for academic demonstrations.
- To showcase the integration of AI and embedded systems in transportation safety.

METHODOLOGY

A camera is installed to monitor the blind spot area of the bus. The camera continuously captures live video frames. Each frame is processed using a machine learning object detection model to detect humans. A predefined danger zone is marked in the camera view. The system checks whether a child is present inside the danger zone. If detected: The LED turns ON to indicate braking action. The buzzer alerts the driver. If no child is detected: The LED and buzzer remain OFF. The system automatically resets when the area becomes clear. This approach simulates braking safely without

FUTURE SCOPE ?

- **Real Brake Integration Connect the output to actual braking systems under controlled industrial conditions .**
- **Multi-Camera Support Cover all blind spots (front, rear, and sides).**
- **Mobile App Alerts Send notifications to drivers or supervisors.**
- **Cloud Data Logging Store incident data for safety analysis.**
- **Advanced AI Models Improve detection accuracy using custom-trained datasets.**
- **IoT Integration Connect multiple buses to a central monitoring system.**
- **Edge AI Optimization Use lightweight models for faster processing on low-power devices**

CONCLUSION

This project presents a safe, intelligent, and practical solution to one of the most critical problems in school transportation safety –child accidents in bus blind spots. By combining AI based human detection with a braking indication mechanism, the system demonstrates how modern vehicles make safety decisions without involving risky hardware control. The prototype successfully shows:

- How AI can prevent accidents
- How embedded systems execute safety logic
- How real-world technologies can be adapted for academic environments

THANK YOU

TEAM TIKI-TAKAS